

Set 3: Random and systematic errors

All measurements have a degree of uncertainty resulting in experimental error. The experimental error in a result is the difference between the experimental value and the literature or theoretical value. There are two types of experimental error: random error and systematic error. Both should be considered when evaluating any quantitative investigation.

Random errors come from measurements that have an equal chance of being above or below the actual value.

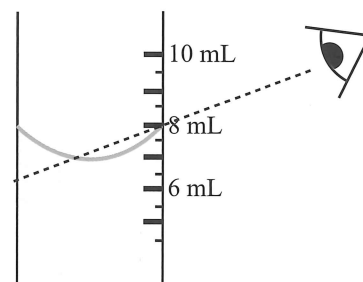
Systematic errors are a result of flaws in the experimental method or apparatus that lead to a result that is always either above or below the true value.

Errors are discussed in detail on page ix and x.

Notes

Set 3: Exercises

- A student measured 8.0 mL of water in a measuring cylinder as shown.
 - What would your reading be?
 - Describe the error/s in the student's technique.
 - Does this error in technique result in random or systematic errors in the volume measurements?
 - What is the impact of this error in technique on the actual volumes measured?
 - What is the uncertainty (measurable random error) associated with reading from this measuring cylinder?
- Digital pH meters are calibrated with solutions of known pH.
 - What kind of error occurs if a solution thought to be pH 7.0 but which is actually pH 7.2 is used in the calibration of the meter.
 - Explain your answer.
- Describe a systematic error that could occur when:
 - measuring the volume of a liquid using a measuring cylinder.
 - measuring the mass of reactants on a top pan balance.
- Describe how random errors result when:
 - Estimating pH using a colour chart following the addition of Universal Indicator to samples of acids and bases.
 - Measuring temperature using a thermometer
 - Judging flame colour when metal salts are heated in a Bunsen flame.
 - Calculating the R_f values (distance the sample migrates/ total distance moved by the solvent.) of samples separated by paper chromatography.
- Describe how it is possible to obtain a set of experimental results that are very similar in value but that are not accurate.
- Describe how the random uncertainty in a measuring the mass of reactants can be minimized.
- A student investigating the rate of a reaction using a digital stopwatch. The results were recorded to one hundredth of a second. Explain why the results are not accurate to one hundredth of a second.



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Notes

8. Repeated chemical analysis of a sample of hematite rich iron ore indicated the percentage by mass of iron present was 56.35%, 56.35%, 56.37%, 56.34% respectively. More accurate analysis then revealed the actual amount of iron present was 57.702%.
- Comment on the reproducibility of the original method and the accuracy of results.
 - Comment on the impact of random and systematic error on the results.
 - How could the accuracy of the results be improved?
9. The energy contained in food can be estimated by burning food and using the energy released to heat up a known mass of water. This experiment can be done with crackers. The sample is ignited using a Bunsen burner and then held below a test tube of water. The temperature rise in the water can be used to estimate the energy in the food.
- Identify variables that should be controlled in this experiment.
 - Explain why the energy content estimated by this technique is always lower than the actual value.
10. Neutralisation reactions are exothermic. The enthalpy change in such reactions can be calculated by working out the heat released into solution. An experiment was carried out adding 100 mL of 1 mol L⁻¹ sodium hydroxide added to 100 mL of 1 mol L⁻¹ hydrochloric acid in a polystyrene coffee cup. The temperature change in the solution is measured using a thermometer and then used along with the mass of solution and specific heat capacity of water to work out the enthalpy change (ΔH). The experimental value calculated in this experiment was only 50 % of the theoretical value.
- Suggest the major source of systematic error in this experiment
 - Identify sources of random error.
 - Suggest improvements that would increase the reliability (validity) of the results.
 - Suggest improvements that would improve the accuracy of the result.
11. Two experiments were carried out to work out the formula of hydrated copper(II) sulfate. In the first experiment (Experiment 1) copper sulfate was added to a crucible and heated for two minutes. Recognising a systematic error in experiment 1 the student carried out a second experiment (Experiment 2) heating an identical sample repeatedly until a constant mass was obtained.
- The measurements made are shown in the table below:

	Time of heating (minutes)	Experiment 1	Experiment 2
		Mass (g)	Mass (g)
crucible		12.20	12.20
hydrated copper sulfate and crucible		13.20	13.20
hydrated copper sulfate		1.00	1.00
anhydrous copper sulfate and crucible	2	12.95	12.94
	3		12.89
	4		12.84
	5		12.84

- State a source of measurable random error in this experiment.
 - Explain why this measurement is considered to have random uncertainty.
 - Assuming there is no uncertainty in taring (zeroing) the balance explain why the overall uncertainty of measurements used to obtain the mass of water in Experiment 1 is ± 0.02 g
 - Describe the systematic error present in the first experimental design.
 - Describe the impact of the error on the number of moles of water (x) in the hydrated formula $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$.
 - Explain how the method in Experiment 2 overcame this systematic error.
 - The student carrying out the second method weighed the sample 4 times. Do the additional measurements increase the uncertainty of the final value they obtain? Explain your answer.
12. The energy released when different fuels undergo combustion can be estimated by burning a known mass of the fuel using the heat released to increase the temperature of water. The temperature change along with other data (shown below) can be used to work out the molar heat of combustion (ΔH).
The results obtained for four different alcohols are shown in the table.

Fuel	Formula of alcohol	Average mass of fuel burned (g)	Average change in temperature of 200 mL of water (K)	Literature value for the molar heat of combustion (kJ mol^{-1})	Experimental value for the molar heat of combustion (kJ mol^{-1})
methanol	CH_3OH	2.40	42	726	385
ethanol	$\text{CH}_3\text{CH}_2\text{OH}$	2.22	45	1367	
propan-1-ol	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	2.64	50	2021	

- Write balanced chemical equations for the complete combustion of each alcohol to water and carbon dioxide.

- Calculate the heat released for the other fuels.
- Calculate the experimental value for the molar heat of combustion for the other fuels.
- Describe the trend in the literature values of molar heat of combustion.
- The experimental values are all lower than the literature values. What major source of systematic error accounts for this difference?
- Calculate the percentage difference between the experimental values obtained in this experiment and the literature values for each of the fuels.

The heat released by each alcohol can be calculated as follows:

$$q = mc\Delta T$$

q = the heat released in joules

m = mass of water heated in g.

The density of water is 1 g mL^{-1} therefore the 200 mL of water heated is converted to 200 g for the calculation.

c is the specific heat capacity of water = $4.18 \text{ J K}^{-1} \text{ g}^{-1}$

Example calculation for methanol

$$q = 200 \times 4.18 \times 42 = 35,112 \text{ J}$$

The molar heat of combustion for each fuel can be calculated using the formula $\Delta H = q/n$

Example calculation for methanol:

$$n(\text{CH}_3\text{OH}) = m/M = 0.0911$$

$$\Delta H = 35,112 / 0.0911 = 385422 \text{ J mol}^{-1} = 385 \text{ kJ mol}^{-1}$$

- Unlike the other two fuels it was observed that the propan-1-ol burned with a sooty flame and left a deposit of black powder on the underside of the flask. Use this evidence to account for the result for propan-1-ol in part f.
- Suggest improvements to the experimental design that could improve the accuracy of the results obtained.